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Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 3, Semester 2 (2024)

IT3060 – Human Computer Interaction

Duration: 2 Hours

November 2024

Instructions to Candidates:

- This paper is preceded by a 10-minute reading period. The supervisor will indicate when answering may commence.
- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 6 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 01 (20 Marks)

Human-Computer Interaction (HCI) is the study and design of how people interact with computers and other digital systems.

a) Identify three (03) advantages of incorporating HCI in application development. **(3 Marks)**

b) Name four (04) UX design principles/factors in HCI. **(2 Marks)**

c) Define the concept of 'Context' in HCI. Briefly explain two (02) out of 5 types of contexts in HCI. **(4 Marks)**

d) State whether you Agree or Disagree with the statements given below. Justify your answer in relation to the user experience (UX) domain. **(6 Marks)**

i. **"Designing for the user experience isn't just about creating beautiful interfaces; it's about crafting interactions that feel human."** - Aarron Walter

ii. **"Usability is about people and how they understand and use things, not about technology."** -Steve Krug

e) Alisia, a co-founder of a rising tech firm, highlights the importance of interacting directly with users and immersing herself in their experiences to guarantee the success of their applications.

i. Explain how 'Design Thinking Process' support Alisia's perspective? **(3 Marks)**

ii. Recommend two (02) HCI related social issues Alisia needs to address when designing the applications. **(2 Marks)**

Question 02 (35 Marks)

Ryan owns a small business called "**Wooden Creations**," where he crafts custom wooden furniture and home décor items. Ryan faces difficulties managing his customer orders because he doesn't have an online store to showcase his products and current prices. Customers either have to call him or visit his workshop to view his designs and inquire about pricing. This process is not only inefficient but also limits his ability to reach a wider audience. Many of his customers have expressed the need for an easier way to browse his products without having to travel to his shop. Ryan also dreams of expanding his customer base internationally, but without an online platform, he finds it hard to showcase his work and scale his business beyond local markets.

Assume you have been assigned to the design team tasked with creating a web and mobile app for "Wooden Creations."

a) The Design Thinking process is a framework that UX designers use to organize their approach to designs. Apply design thinking model for the above scenario and briefly describe each phase. " **(10 Marks)**

b) Identify three (03) user groups for the "Wooden Creations" app and provide a brief description of each group. **(4.5 Marks)**

c) Select two (02) user groups and create hypothetical personas for each group. Use the template below to complete the personas. **(6 Marks)**

Picture	Demographics
Behaviors	Goals & pain points

d) User Interviews have always been a popular choice for user research. Identify three (03) types of interviews and compare each by highlighting two (02) distinct aspects.

(6 Marks)

e) Recommend three (03) user research techniques, excluding those listed in part (d), that are appropriate for launching the "Wooden Creations" app on the online market. Justify your recommendations. **(6 Marks)**

f) Ryan has chosen to utilize social media for marketing until the launch of the "Wooden Creations" app.

"Social media platforms are widely popular due to the emotional reactions they generate." Do you agree or disagree with this statement? Justify your answer using sciences in HCI. **(2.5 Marks)**

Question 03 (30 marks)

"**FitTrack**" will offer professionals a simple and efficient platform to plan their fitness routines for both weekdays and weekends. With just a few taps on their smartphones, professionals can schedule workout sessions, choose from a variety of exercise options such as gym workouts, yoga classes, or outdoor activities, all available through local fitness centers and trainers listed within the app. They can personalize their fitness plans based on their goals, preferences, and fitness levels, ensuring a routine that fits their busy work schedules and health objectives.

Assume that you have been appointed as the UI/UX engineer to design an application for the scenario mentioned above.

a) Explain two (02) ideation techniques you would apply for the given scenario. **(4 marks)**

b) Analyze the given scenario and identify two (02) functional requirements and non-functional requirements. **(4 Marks)**

c) Analyze the above scenario and identify two (02) possible accessibility issues /disabilities expecting as an UX designer. Recommend design suggestions for the identified accessibility issues. **(4 Marks)**

d) Explain the importance of applying participatory design to the above scenario. **(3 Marks)**

e) Analyze the above scenario and explain two (02) design principles to improve the user experience in the " FitTrack" application. **(4 Marks)**

f) As a UI/UX engineer you are responsible for creating prototypes.

i. Briefly explain two (02) major variations of prototypes. **(4 Marks)**

ii. **"Paper prototyping is a process where design teams create paper representations of digital products to help them realize concepts and test designs."** Do you recommend a paper prototype for the "FitTrack" application. Justify your answer by providing two (02) advantages of paper prototypes. **(3 Marks)**

iii. From the two variations presented in part (f) (i), recommend which variation allows management to provide constructive feedback on both interactions and the overall system architecture. Justify your recommendation by providing two (02) examples for the selected variation. **(4 Marks)**

Question 04 (15 Marks)

a) Explain two (02) out of 5 dimensions of interaction. **(3 Marks)**

b) Explain two (02) types of interactions you would incorporate when designing applications. **(2 Marks)**

c) "Heuristic Evaluation is an appropriate approach for assessing the usability of an interface." Explain what is mean by Heuristic Evaluation. **(3 Marks)**

d) Differentiate formative evaluation and Summative evaluation. **(4 Marks)**

e) Explain what the 'Think Aloud" Evaluation method is and its significance. **(3 Marks)**